STORIES SHAPE SOLUTIONS: BALANCING HUMAN-BEAVER RELATIONSHIPS TO BUILD RESILIENT WATERSHEDS

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Beaver dam in the Flat Tops Wilderness, CO Photo Credit: Becky Edmiston

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Abstract

Human perspectives play a critical role in shaping the success of ecological restoration efforts, particularly in systems where wildlife and working landscapes intersect. This study uses a transdisciplinary research approach to examine stakeholder experiences with beavers in the Upper Yampa Basin. Qualitative data was collected through 18 semi structured interviews with ranchers, landowners, and local stakeholders and analyzed using narrative based thematic coding. Findings show strong agreement across stakeholder groups on the ecological benefits of beavers, particularly for water storage, that contributes to drought and wildfire resilience, while also revealing that lived experiences shape how these benefits are understood in practice, with landowners emphasizing localized challenges and professionals focusing on watershed scale processes. Social dynamics, including trust, relationships, and exposure to local solutions, emerged as central factors shaping coexistence outcomes. While the ecological benefits of beavers are widely recognized, it is these personal stories that influence whether they are seen as a problem or a partner, highlighting that successful restoration depends on aligning ecological knowledge with community perspectives through place based and collaborative approaches.

Keywords: beaver, coexistence, watershed resilience, stakeholder perspectives, human-wildlife relationships, narrative inquiry, restoration, sustainability, agriculture, rancher, education, biodiversity, story-telling, Upper Yampa Basin, Colorado, drought, wildfire.

Introduction

Across the world, rivers are drying up, wildfires are spreading, and ecosystems are deteriorating. Human interaction with the land has caused harm through centuries of overuse and extraction. Industrial farming and resource exploitation have stripped once fertile soil to dust, years of burning fossil fuels have filled the atmosphere with pollution, and economic systems built on constant growth have disrupted the balance that once kept Earth stable. Now, droughts last longer, storms grow more unpredictable and extreme, glaciers are melting, and water has become a scarce resource. Climate change is the visible outcome of centuries of human impact, a global crisis that exposes how the natural systems supporting life are reaching their breaking point.

Sustainability offers a framework for confronting this crisis. We must learn how to live within those limits and create resilient systems that can endure long into the future. Over the past fifty years, scientists, researchers, and global leaders have worked to understand how humans can live on a finite planet without destroying it. Early studies warned that endless growth would lead to collapse. Later, global reports and scientific models built a clearer vision of what balance between people and the planet could look like. These ideas together shape the foundation for sustainability today.

The first major warning came from *The Limits to Growth: a Report for The Club of Rome's Project on the Predicament of Mankind* (1972), published by The Club of Rome, a diverse group of scientists, economists, and business leaders founded in 1968 to explore the long-term future of humanity and drive policy change (Meadows et al., 1972, p. 9). Using one of the earliest computer models of global systems, experts at MIT revealed how unchecked population and industrial growth could exceed the planet's capacity to sustain life. Thankfully,

what also emerged is that "It is possible to alter these growth trends and to establish a condition of ecological and economic stability that is sustainable far into the future." (Meadows et al., 1972, p. 24). These findings shocked the world, sparking environmental movements and fierce debate about the limits of economic growth. This was the first time scientists used systems thinking to show that the planet's resources are not infinite and that human activity must stay in balance with nature to survive.

Fifteen years later, the United Nations *Report of the World Commission on Environment and Development* released *Our Common Future* (1987), more commonly known as the Brundtland Report, which gave sustainability a holistic and ethical meaning. It defined sustainable development as "meeting the needs and aspirations of the present without compromising the ability to meet those of the future" and emphasized the importance of environmental protection (WCED, 1987, p. 39). The report stated that "sustainability requires views of human needs and well-being that incorporate such non-economic variables as education and health enjoyed for their own sake, clean air and water, and the protection of natural beauty" (WCED, 1987, p. 49). It also called for strong laws to protect "an environment adequate for health and well-being" (WCED, 1987, p. 56). The report illuminated that the health of the environment, the strength of the economy, and the well-being of people are all deeply connected. This interdependence laid the conceptual groundwork for the three foundational pillars of sustainability: environment, economy, and equity, showing that lasting progress depends on keeping them in balance.

Donella Meadows, one of the authors of *The Limits to Growth*, expanded on these ideas in *Thinking in Systems* (2008). Her work helped people see the world as a web of connections. She defines a system as "an interconnected set of elements that is coherently organized in a way

that achieves something (Meadows, 2008, p. 11). She explained that problems such as climate change, biodiversity loss, and pollution are not separate, as they are linked through feedback loops that influence one another. Meadows reminded us that the world is a dynamic and complex system that does not behave linearly. Her book describes how systems can behave in unexpected ways, where small changes ripple outward and create larger effects, while others disappear with no visible impact. She described key ideas like resilience, leverage points, and boundaries. Systems thinking helps us recognize that everything is connected, and when one part of the system fails, the rest begins to follow. She called these leverage points, or the places where thoughtful action can shift an entire system. She also described the limits that keep systems stable, showing that when those boundaries are crossed, systems begin to break down. Meadows pointed out traps such as the "tragedy of the commons," where individuals acting in self-interest exhaust shared resources, and she emphasized the importance of resilience, or the ability of a system to recover after disturbance. Together, these lessons reveal that sustainability is not about controlling nature but about understanding its complexity and learning to work within it.

Building on these foundations, scientists at the Stockholm Resilience Centre introduced the *Planetary Boundaries* framework in 2009 (SRC, 2025, para. 2). Today, this model identifies nine environmental systems that keep the planet stable, including changes in the climate, land, and water. It warns that these boundaries are connected: "The long-term stability of the past, which allowed human societies to develop and thrive, comes from the complex interactions of biophysical processes within the Earth system" (SRC, 2025, para. 5). The *Planetary Boundaries* urge us to consider this independent relationship when making decisions about the future of humanity and the rest of life on Earth. Every year, the boundaries are being pushed further and further past their limits, with seven out of the nine boundaries being crossed in 2025. As the

world moves beyond its safe operating space, the challenge becomes how to rebuild balance by focusing on the places and species that still hold the power to restore it.

The Upper Yampa Basin in northwestern Colorado is one small place where this global crisis is coming to life. Snowpack is shrinking, rivers are running dry, and wildfires are becoming far too frequent. According to the National Park Service (2015), "The Yampa is the only remaining free-flowing tributary in the Colorado River system," meaning it is not regulated by major dams along its main channel and still retains a relatively natural flow regime. Its hydrograph, the natural rise and fall of flows, acts as a heartbeat, pulsing water through the system and sustaining surrounding ecosystems. This makes the river a vital lifeline for the landscape and a unique opportunity to study how natural processes contribute to resilience. The Yampa River reflects the broader sustainability challenge of how humans live within natural limits. As the river changes, so do the communities that depend on it. The problem is complex, yet the solutions are achievable when people approach them systematically.

One surprising source of hope in this landscape comes from an old ally: the beaver. Once nearly wiped out by fur trapping, beavers are returning and quietly repairing what was lost. Their dams slow water, spread it across floodplains, and create wetlands that store carbon and provide habitat for countless species. These ponds act like natural sponges, keeping the landscape green during droughts, and can even act as natural firebreaks. By building and maintaining wetlands, beavers turn destructive feedback loops into stabilizing ones. Their work embodies the principles of resilience and adaptation that sustainability requires.

Coexistence with beavers, however, depends on human attitudes and cooperation. Some ranchers and landowners see them as partners who improve soil health and water storage. Others view them as pests that flood fields or block culverts. These different perspectives shape

restoration efforts and determine whether people choose to work with or against nature. This research connects global sustainability theory to emerging local understanding, exploring these perceptions through a sustainability lens

This capstone project, *Stories Shape Solutions: Balancing Human-Beaver Relationships to Build Resilient Watersheds*, explores how local attitudes shape opportunities for beaver-based restoration. Beavers show how the three pillars of sustainability: environment, economy, and equity, interact through the story of one keystone species. Environmentally, beavers create resilience, improving ecosystems and enhancing biodiversity. Economically, they provide ecosystem services that save time and resources. Socially, they challenge people to rethink their relationship with nature and inspire communities to work collectively toward regenerating healthy ecosystems and relationships.

The sustainability crisis asks the same question everywhere: how can we adapt to a changing climate and build resilience into the future. It takes a holistic understanding of the systems that we are all a part of. The lessons from *The Limits to Growth*, *The Brundtland Report*, *Thinking in Systems*, and *Planetary Boundaries* suggest that we can, if we learn to see ourselves as part of the larger system. The Yampa Valley demonstrates how these ideas can take root in a real place, reminding us that lasting change begins within local landscapes and communities. If people can learn to coexist with beavers, they may also learn how to live more sustainably within the boundaries of the Earth itself.

Literature Review

Understanding the story of human-beaver coexistence reveals more than ecological history; it reflects humanity's broader relationship with the living world. The literature shows that attitudes toward beavers have shifted dramatically over time from sacred kin to exploited resource, and finally to ecological partner. These changes mirror the evolution of sustainability itself, moving from reciprocity with the living world, to dominance and extraction, and then back toward balance and harmony. To understand how local perceptions in the Yampa Valley influence beaver-based restoration, it is important to trace both the ecological science and the social meaning behind this species. The existing research spans ecology, economics, and society, and together they form a story of how people are learning once again to live within natural limits.

Historical and Indigenous Perspectives

Long before the word sustainability existed, Indigenous people across North America practiced it. Among the Nishnabek tribe of the Great Lakes, the beaver is honored as a wise relative: "The elders tell us that Mek, the four-legged fur bearer we call beaver, is our eldest teacher" (Garrity, 2024, para. 1). Jessica Garrity, a member of the Nishnabek nation, explains that this teaching is central to her people's understanding of who they are and where they come from: "In this story, the beaver is our first teacher, and we learn how to utilize tools available to us for safe passage" (Garrity, 2024, para. 3). In many Indigenous nations, beavers are viewed not as resources but as relatives who model balance, patience, and resilience.

In the ancient Cheyenne story "The Gnawing," the Great White Grandfather Beaver is said to hold up the world while gnawing at the pole of creation. When he grows angry, he gnaws faster, threatening to bring the pole and the world itself crashing down. Because of this, the people never harm or eat beavers, knowing that their fate depends on keeping him calm (Erdoes

& Ortiz, 1984, p. 485). The story teaches deep respect for the beaver's power and reminds listeners that balance and survival depend on honoring the beings who sustain the world. This ancient wisdom reveals that Indigenous peoples have long understood sustainability as a lived practice rooted in relationship. Their ways of knowing emphasize reciprocity and responsibility, as the belief that caring for the land and water also means caring for one's community. Together, these oral traditions and teachings frame the beaver not simply as an animal, but as a guardian of balance, showing that sustainability begins with trust and respect, which are lessons that continue to shape Indigenous stewardship today.

This knowledge continues to guide modern Indigenous stewardship. In "Beaver Believers: Native Americans Promote Resurgence of 'Nature's Engineers,'" journalist Louise Sherriff highlights how tribes across the West are restoring ancestral relationships with beavers as part of both climate and cultural resilience. One tribal member explained, "They've gone through the same struggles against Europeans as us. The beavers were viewed as pests and driven off the land. We're looking to bring the beaver back again, to help us manage the land like they used to" (Sherriff, 2021, p. 3). This movement reflects an enduring worldview that the beaver, as both elder and engineer, can help heal what colonial exploitation has damaged. Indigenous-led efforts reconnect cultural identity with ecological restoration, showing that sustainability is as much about relationship and respect as it is about science.

The balance between humans and beavers shifted dramatically with European colonization and the rise of the fur trade. Beavers, once regarded as kin, became one of the most sought-after commodities in the world. Long before colonists arrived in North America, European trappers had already hunted their own continent's beavers (*Castor fiber*) to near extinction for the procurement of luxurious pelts for fancy fur hats, and castoreum used in

perfume and medicine (Goldfarb & Flores, 2018; Fairfax and Westbrook, 2024, p. 328). When that resource vanished, traders turned to North America, where rivers teemed with *Castor canadensis*. From the 1600s through the 1800s, the booming fur industry drove a relentless hunt that reshaped entire landscapes. As Ben Goldfarb explains in *Eager,* "with breathtaking speed, the mountain men demolished their resource, virtually wiping out beavers throughout the American West" (Goldfarb & Flores, 2018, p. 4). Millions were killed, and their disappearance drained wetlands, altered hydrology, and devastated biodiversity across the continent. What had once been vibrant, water-rich landscapes turned dry and degraded (Goldfarb & Flores, 2018, p. 9). The near eradication of beavers disrupted entire watersheds, showing how quickly human economies can collapse the ecological systems that sustain them.

It took more than a century for reintroduction and conservation efforts to begin reversing this loss. By the early 1900s, wildlife managers started translocating beavers to restore water storage, prevent erosion, and stabilize stream systems (Pollock et al., 2023, p. 22). Decades later, as land use and grazing practices improved, beavers began to recolonize on their own. In Elko County, Nevada, for instance, riparian recovery in the 1990s led to the natural return of beavers (Charnley, 2019). As one rancher described, "My whole attitude with beavers has changed in this 20 years… I think they're going to be one of the keys to restoration of western streams" (Charnley, 2019, p. 30). This shift marked a broader transformation in public perception: from viewing beavers as pests to recognizing them as partners in rebuilding ecological resilience.

Environmental Dimension: Beavers as Ecosystem Engineers

Scientific literature consistently describes beavers as keystone species whose work enhances entire ecosystems. Their dams slow water, store sediment, and create wetlands that serve as nurseries for fish, birds, and amphibians (Brazier et al., 2021). These wetlands filter

pollution, increase groundwater recharge, and provide drought resilience. Research shows that beaver ponds raise the water table and keep landscapes green even during wildfires (Fairfax & Whittle, 2020). By spreading and storing water, beavers help regulate the extremes of flood and drought, which are two of the most pressing consequences of climate change.

Building on this foundation, Jordan and Fairfax describe beavers as a solution in "the North American freshwater climate action plan," emphasizing their ability to sequester carbon and restore natural hydrology. The science shows "beaver-based restoration creates profound opportunities for initiating positive feedback cycles and increasing riverscape resilience" (Jordan and Fairfax, 2022, p. 2), a real-world example of the systems thinking Donella Meadows described. The Beaver Restoration Guidebook (Pollock et al., 2023) and similar resources like Tippie's (2010) *Working with Beaver for Better Habitat Naturally* document these benefits and offer practical tools for implementation. Their ponds increase resilience by capturing nutrients and slowing runoff, creating self-sustaining systems that require little human intervention. In this way, beaver restoration offers a low-tech, process-based approach to climate adaptation, aligning perfectly with the environmental pillar of sustainability.

Furthermore, a regional study by wildlife biologists in Colorado compared beaver populations on the regulated Green River and the free-flowing Yampa, using radiotelemetry and GIS mapping to track colony density and habitat use (Breck et al., 2001). They found that beavers on the Green thrived where steady flows supported willow growth, while beavers on the Yampa had to travel farther from the stream to find food, using more energy and facing higher risks from predators. The findings challenge the belief that nature is always best left alone, suggesting that in places already shaped by human influence, thoughtful management can sometimes restore balance. This study offers a philosophy to guide future work, reminding us

that restoration begins when we ask not only what beavers can do for us, but what we can do for them.

Economic Dimension: Costs, Benefits, and Coexistence

In contrast, some researchers flip this question around, exploring the tangible ways that beavers support human well-being. The article "Ecosystem Services Provided by Beavers Castor spp." explores this utilitarian viewpoint, focusing solely on the benefits of nature for humans. Using existing ecological and economic data from across the Northern Hemisphere, the authors modeled how beaver activity translates into monetary value, estimating the worth of their flood control, carbon storage, and water purification services. They found that these benefits generate tens of millions of dollars annually and suggest programs such as Payment for Ecosystem Services to reward landowners who support coexistence (Thompson et al., 2021, p. 36). The authors also note that maintaining wetland health is a public concern, yet much of this habitat lies on private land (Thompson et al., 2021, p. 34). This tension mirrors challenges in the Yampa Valley, where successful restoration depends on cooperation between individual landowners and the broader community. While the focus is economic, the study shows that even a human-centered approach can expand how decision-makers think about conservation, helping societies regulate and utilize beaver activity in ways that improve well-being and encourage greater stewardship of the natural world.

While beavers provide immense ecological benefits and economic value, they also challenge human infrastructure. The literature on economics and management reflects this tension. In their study, *Mitigating Infrastructure Loss from Beaver Flooding: A Cost–Benefit Analysis,* the authors compared traditional methods of trapping and dam removal with adaptive management solutions such as flow devices and culvert protectors. They found that coexistence

strategies saved money over time and reduced recurring damage, although they mentioned the importance of combining cost-benefit analysis with on-the-ground testing to inform more effective management of natural resources (Hood et al., 2018, p. 158). Manuals such as Wheaton's *Low-Tech Process-Based Restoration of Riverscapes* (2019) and Hodge's *Process-Based Restoration Monitoring Handbook* (2025) expand on this idea, framing beavers as natural collaborators whose work reduces the need for expensive engineering. These projects rely on inexpensive, natural materials and often succeed because beavers finish the job by maintaining the structures over time.

Jordan and Fairfax (2022) also link beaver restoration to cost savings in climate adaptation, noting that natural hydrologic repair is cheaper and longer lasting than human-built interventions. After all, beavers do the work for free! Research by Charnley (2019) found that ranchers who once opposed beavers now see them as assets for improving water retention and forage productivity. This shift highlights an essential point within the economic pillar of sustainability: long-term prosperity depends on working with, rather than against, natural processes.

Equity and Human Dimensions: Coexistence and Collaboration

In a different study, *Ranchers and Beavers: Understanding the Human Dimensions of Beaver-Related Stream Restoration on Western Rangelands,* Charnley and her team emphasize that successful restoration depends on relationships among people as much as on ecological conditions. Some landowners view beavers as nuisances that clog culverts and flood roads, while others see them as symbols of renewal. Their findings show that perceptions are shaped by experience, trust, and local knowledge. As the authors write, "Social and cultural values shape restoration goals and actions, and the ecosystem services that people desire inform restoration

interventions" (Charnley et al., 2020, p. 4). This shows that restoration is not just about the physical work of moving water and sediment, but also about aligning management goals with the values of those who live on the land.

In his article, *Beavers Rebooted,* Ben Goldfarb notes that even as beaver restoration gains traction through beaver dam analogs [BDA's], it continues to face resistance. Many regulators remain cautious about the method, and government agencies still kill thousands of beavers each year that are labeled as nuisances (Goldfarb, 2018, p. 158). This lingering distrust shows that scientific understanding alone cannot ensure coexistence; social acceptance is equally vital.

Hood and Yarmey (2015) expanded this perspective through an adaptive management project that combined mapping with public engagement. Using GIS to locate more than 350 beaver lodges in Alberta, they identified conflict hotspots and found that inexpensive pond-leveler devices reduced flooding and complaints. Their work illustrates that technical solutions succeed only when paired with communication and trust between managers and the community.

Building on this perspective, Moravek and her colleagues expand on the importance of dialogue between local communities and scientists. Their study, *Using Beaver Capacity Models: The Importance of Local Knowledge,* was designed to bridge these groups. As they explain, "We intend for this study to create an open dialogue between scientists, communities, and land managers about how beaver models can be applied most effectively to inform beaver-related restoration" (Moravek et al., 2025, p. 1). This intention, to unite scientific modeling with local understanding, reflects the heart of the equity pillar of sustainability. It ensures that decisions about restoration are informed not only by data but by the lived experience of those who share the landscape with beavers.

Practical guides such as Vorel's *Handbook for Coexisting with Beavers* (2016), Hawley-Yan's *Techniques for Mitigating Human/Beaver Conflicts in Urban and Suburban Environments* (2016), and *Beavers: Coexistence Strategies for Municipalities and Landowners,* published by The Fur-Bearers (2019), further contribute to this dialogue by offering coexistence strategies for municipalities, landowners, and agencies. Together, these resources highlight the importance of shared stewardship and show how collaboration among communities, land managers, and policymakers can make coexistence possible.

Synthesis and Research Gap

Across the literature, beavers emerge as environmental engineers, economic allies, and cultural teachers. Scientific studies confirm their role in climate resilience, while management research highlights the financial logic of coexistence. Social studies reveal that success depends on trust, communication, and shared responsibility. Together, these works show that researchers are increasingly bridging the three pillars of sustainability through collaborative and interdisciplinary approaches. However, gaps remain in ground-truthing and local observation. Much of what is known comes from models or broad-scale studies, while field-based data about what materials beavers use, where they build, and how they adapt to different habitats is still limited. Beavers are generalists, capable of using everything from willows and aspens to cattails and mud, and their adaptability varies from place to place. Ranchers and other local land stewards hold place-specific perspectives that are essential for understanding how beaver-based restoration can be effectively implemented in the Yampa Valley. To strengthen this growing field, more research is needed to connect broad scientific knowledge with on-the-ground realities and community experience, which is what I aim to achieve with my capstone project.

The story of the beaver mirrors the story of sustainability itself. From sacred teacher to overexploited resource to ecological ally, the beaver reflects humanity's shifting relationship with nature. The literature reveals that beavers not only revitalize wetlands and stabilize rivers but also invite people to restore their sense of belonging within the natural world. Understanding this relationship requires looking beyond biology to include economics, equity, and cultural meaning. In doing so, we see that coexistence with beavers is more than a conservation goal, but it is a model for how people can once again live in partnership with the systems that sustain life.

Methods

This study employed a transdisciplinary, mixed methods approach to examine how human–beaver relationships influence watershed resilience in the Upper Yampa Basin. Transdisciplinarity requires working across boundaries by integrating ecological science with local knowledge through engagement with a range of stakeholders, while valuing all perspectives equally. Mindful inquiry emphasizes slowing down, questioning assumptions, and treating participants as co-interpreters of meaning rather than subjects of a study.

Participants were approached with curiosity, compassion, and collaboration, with an emphasis on listening to their stories. This allowed for a deeper understanding of how lived experiences shape perspectives on beaver management and coexistence. Together, these approaches are essential for a project that examines human perspectives on beavers and restoration in the Yampa Valley, where ecological processes and social worlds are tightly interconnected.

The study was conducted in the Upper Yampa Basin in northwestern Colorado. The region includes a mix of private agricultural lands and public lands with varying levels of beaver activity (Figure 1).

Data Collection

Qualitative data was collected through 18 semi structured interviews, which invited participants to tell stories, describe experiences, and reflect on the meanings they attach to beaver presence or conflict. Interviews are not neutral data extraction events but social encounters where meaning is co-constructed, describing narratives as a way people make sense of their world by organizing events, emotions, and consequences over time (Marvasti, 2019). Narrative inquiry is, therefore, an appropriate framework for understanding how ranchers, landowners, and local

stakeholders construct and communicate their perspectives on beavers as both problem and partner. Chapter 5 highlights that democratizing knowledge production begins with recognizing community members as experts on the places they inhabit. This approach also reflects the principle that knowledge is co-produced, recognizing community members as experts on the places they inhabit.

Semi-structured Interviews

Interviews followed a semi structured format, allowing for consistency across participants while also providing flexibility to explore individual experiences in depth. Open questions encouraged participants to share personal stories, describe challenges or benefits, express their attitudes toward coexistence, and reflect on how their views of beavers have changed over time. This flexible format allowed participants to guide the conversation while keeping the discussion centered on the research question.

The interviews lasted anywhere from 30-90 minutes, ten of which were conducted in person and eight were conducted over the phone. All participants signed an informed consent form in accordance with DHHS ethical research standards. Interviews were audio recorded, and brief field notes were also taken during each interview.

Sampling and Participants

Participants were recruited through existing professional networks and referrals using purposive and snowball sampling (Marvasti & Treviño, 2019, p. 34). Initial stakeholders were intentionally selected based on their relevance to the research questions. Subsequent participants were identified through recommendations from these initial contacts, allowing the sample to expand.

Two key stakeholder groups were identified. The first group consisted of ranchers and landowners with firsthand experience of beaver presence, conflict, or benefits on their land. The second group included local stakeholders such as restoration professionals, wildlife biologists, nonprofit organizations, government agencies, and educators.

Interviews were conducted from January - March 2026. 10 in person, really valuable. Getting to visit ranches, connect in small ways, like being welcomed with a batch of cookies. and offices. 8 over the phone, still in depth and meaningful

Community Events

In addition to individual interviews, participation in regional events provided broader context and exposure to diverse perspectives. These events included the Yampa Basin Rendezvous, the Colorado Beaver Gathering, the Yampa Valley Beaver Working Group, and the State of the Colorado River Meeting.

The Yampa Basin Rendezvous, attended in May 2025, served as an early influence on the direction of this project. The event focused on watershed resilience and brought together a range of local stakeholders, highlighting both the ecological importance of beavers and the complexity of implementing restoration in working landscapes. This experience helped shape the research focus on balancing human and beaver relationships within a real-world context.

The Colorado Beaver Gathering, held in Boulder, brought together participants from across the country and provided exposure to national-scale conversations around beaver restoration. In contrast, the Yampa Valley Beaver Working Group emphasized local collaboration, where stakeholders discussed management strategies, funding opportunities, educational initiatives, and coexistence approaches specific to the region. The State of the

Colorado River Meeting further expanded understanding of broader water management challenges and regional dynamics.

Together, these experiences helped connect large-scale restoration efforts with local, place-based realities. They also highlighted the importance of collaboration across stakeholder groups and reinforced that successful beaver management depends on aligning broader initiatives with the specific needs and perspectives of local communities.

Spatial Context

Existing spatial datasets, including the Beaver Restoration Assessment Tool and the Colorado Beaver Activity Mapper, were used to provide contextual understanding of beaver distribution and habitat potential in the Upper Yampa Basin. These datasets supported interpretation of interview data by offering a broader ecological perspective, but they were not formally analyzed as part of this study.

Qualitative Analysis

All interviews were transcribed and analyzed using narrative-based thematic analysis. Each transcript was first read in full to understand the overall narrative and how participants made meaning of their experiences. Codes were then developed to identify recurring themes related to conflict, benefit, management practices, values, emotions, and changes in perspective. Coding was conducted using Delve software, following a multi step process. Initial open coding used to identify recurring themes related to conflict, benefit, management practices, values, emotions, or notable turning points in perspective. These codes were then grouped into broader themes, which were further organized across the three dimensions of sustainability: social, economic, and environmental.

Consistent with narrative inquiry, analysis also considered metaphors, storylines, and the broader social and ecological context referenced by participants. Codes were refined iteratively as patterns emerged, allowing themes to remain grounded in participant perspectives, rather than imposing external biases.

Limitations

Every method has constraints, and understanding these limitations is essential for mindful inquiry. Interview-based research relies on participants' memory, interpretation, and comfort with the researcher, which may influence what they choose to share or how they frame their experiences. Critical social science also highlights that the researcher's background in sustainability and conservation may shape interpretation, and that differences in worldview or perceived power dynamics may affect how openly participants discuss beaver-related challenges.

Additional limitations were related to access to transcription and coding tools. Many transcription software platforms require paid subscriptions to process longer audio files, and free versions often impose strict time limits. Although Colorado Mountain College provides access to Microsoft Word's transcription service, usage is capped at 300 minutes per month under the basic plan. As a result, transcription had to be spread across multiple months to remain within these limits, which slowed the analysis process and introduced delays between data collection and coding.

Findings

Balancing Benefits and Challenges

Participants consistently described beavers through a balance of benefits and challenges rather than as strictly positive or negative (Figure 2). Ecological benefits such as water storage, habitat creation, and drought resilience were widely recognized. Landowners also described challenges including irrigation impacts, culvert blockage, and flooding.

Differences across stakeholder groups reflected variation in lived experience rather than disagreement. Ranchers emphasized site specific challenges tied to daily land management, while conservation professionals focused on broader watershed scale processes.

Language and Framing

A word frequency analysis was conducted to examine how participants described beavers in their own words (Figure 3). Only participant responses were included to avoid bias introduced by repeated interview prompts. The resulting word cloud highlights commonly used terms related to both ecological processes and management concerns, including water, habitat, and infrastructure. Participant language rarely framed beavers in absolute terms. Instead, descriptions emphasized balance, tradeoffs, and context.

Environmental Findings

Participants consistently identified environmental benefits of beavers, particularly their role in improving watershed function. These benefits included slowing, storing, and spreading water, increasing biodiversity, and enhancing drought and wildfire resilience (Figure 2). Agreement on these functions was widespread across stakeholder groups.

Many participants described these outcomes through direct observation. One landowner explained, “After I saw that… I can see how the beavers could make a positive impact and prevent erosion… I really benefited from that.” This reflects how ecological understanding is often shaped through lived experience.

At the same time, participants noted environmental tradeoffs. For example, one participant shared, “They've killed some of the cottonwood trees, which makes me very sad.” This highlights that environmental outcomes are not universally positive, but vary by location and context, with benefits often occurring alongside localized impacts.

Economic Findings

Economic impacts were among the most immediate and tangible concerns, particularly for ranchers and landowners. Participants emphasized that beaver activity can interfere with infrastructure, leading to flooding of roads or fields and disruption of irrigation systems (Figure 2). One rancher described this tension clearly: “Beavers are frustrating and impressive at the same time. They get into our irrigation ditches and that's where they're frustrating.” These impacts often translate into increased labor demands and financial costs.

Participants also identified economic opportunities associated with beavers. Nature based solutions were often viewed as more affordable than traditional engineering approaches, and some noted that improved land health can enhance long term value. Increased recognition of sustainable practices may also contribute to consumer support for ranchers. This reveals a key pattern: economic costs are immediate and localized, while economic benefits tend to emerge over longer timeframes and at broader scales.

Social Findings

Social dynamics emerged as a central factor shaping how participants understand and respond to beavers. Key challenges included mistrust, difficulty reaching rural landowners, and conflict where beaver activity overlaps with human infrastructure (Figure 2).

Participants consistently emphasized the complexity of beaver related issues. As one participant explained, “Two things can be true at once… beavers can be really beneficial to the environment and really detrimental to ranchers… there's a lot of nuance to the beaver issue.” This reflects a broader pattern where individuals hold both positive and negative views simultaneously.

Education and outreach were identified as important opportunities, particularly when grounded in local relationships. Participants emphasized that peer learning is more effective than top down approaches. For example, one participant noted, “Once one neighbor does something, then it kind of catches on,” while another explained, “Setting an example… works a lot better than an outsider coming in and saying, this is what you got to do.”

Participants also highlighted the importance of seeing real world examples, stating, “It'd be informative to see other people's property.” These findings suggest that trust, relationships, and lived experience are critical to successful coexistence. Social acceptance is shaped through observation, collaboration, and local credibility rather than information alone.

Synthesis of Findings

Across environmental, economic, and social dimensions, a consistent pattern emerged (Figure 2). Environmental benefits are widely recognized, economic challenges are immediate and influence tolerance, and social dynamics determine how these tradeoffs are navigated. While

the ecological benefits of beavers are widely recognized, it is these lived experiences and social relationships that ultimately shape whether they are seen as a problem or a partner.

Discussion

The findings largely align with initial expectations. As anticipated, participants expressed a wide range of perspectives shaped by land use, location, and personal experience. Most participants recognized the ecological benefits of beavers, while also describing real challenges related to flooding, infrastructure, and land management. Therefore, this sample reflected more supportive than skeptical perspectives, though support was often conditional and dependent on individual experience.

However, the level of agreement across stakeholder groups was stronger than expected. Differences in perspective were less about whether beavers are beneficial and more about how their impacts are experienced and managed. This reinforces that coexistence is not defined by opposition, but by context. Perspectives also differed between the general public and those directly managing land, with landowners emphasizing practical, day-to-day challenges more strongly.

Complex & Interdependent System

These findings highlight that beavers and humans exist within a complex and interdependent system. Beaver activity influences hydrology, vegetation, and ecosystem processes, while human decisions shape where and how those impacts are tolerated. These relationships operate as a dynamic and non-linear system, where small changes in either ecological conditions or human decisions can lead to varied and unpredictable outcomes.

Beavers function as a leverage point within watershed systems. By slowing, storing, and spreading water, they create feedback loops that increase ecological resilience. However, the outcomes of these processes are not determined by ecology alone. They are shaped by how people interact with and respond to those changes. This reinforces that restoration is not solely an environmental process. It is also social, requiring attention to human values, land use, and lived experience.

The Role of Lived Experience and Conditional Support

A key finding is that support for beavers is conditional and shaped by personal experience. Participants who had direct positive interactions with beavers were more likely to view them as beneficial, while those who experienced infrastructure challenges were more cautious or resistant.

This suggests that perceptions are not fixed, but evolve over time through interaction with the landscape. For some, perspectives are shaped by generational knowledge and long-standing challenges. For others, they emerge from recent exposure to restoration efforts or observed ecological benefits. Cultural values and identity also play a role, particularly for those with long-standing relationships to the land and deep ties to working landscapes.As a result, coexistence cannot be treated as a one-size-fits-all solution. It varies by location and depends on both ecological conditions and individual experience.

Learning Through Relationships and Experience

Findings show that people learn about beavers primarily through relationships, observation, and shared experience rather than through formal education alone. Participants emphasized the importance of neighbors, local examples, and real-world demonstrations in shaping their understanding. This challenges the assumption that increasing education alone will

lead to improved coexistence. Information is necessary, but not sufficient. Trust, credibility, and local relevance determine whether knowledge is accepted and applied. These findings highlight the importance of peer-to-peer learning and place-based engagement. Strategies grounded in local relationships are more likely to succeed than top-down approaches.

Implications for Sustainability and Restoration

These results align with the three pillars of sustainability, where environmental benefits must be balanced with economic realities and social acceptance. While ecological functions of beavers are widely understood, successful restoration depends on aligning those benefits with the needs and values of local communities. Economic challenges, particularly related to infrastructure and labor, influence tolerance for beaver activity. At the same time, social dynamics determine how those challenges are addressed. Restoration outcomes are therefore shaped by the interaction of environmental, economic, and social factors rather than any single dimension. Economic benefits were more difficult to quantify and often less visible, while costs were immediate and experienced directly by landowners. This reinforces that long-term success requires listening to stakeholder perspectives, building trust, and developing solutions that are both ecologically effective and socially acceptable.

Together, these findings show that beaver coexistence is not simply a technical problem to be solved. It is a dynamic, place-based process shaped by relationships between people, land, and water. Listening to all stakeholder perspectives is essential for effective, long-term success. Coexistence depends on recognizing complexity, adapting to local conditions, and supporting collaboration across different experiences and priorities.

Conclusion

This study shows that people understand and relate to beavers through stories shaped by lived experience, place, and daily interaction with the land. While the ecological benefits of beavers are widely recognized, it is these personal experiences that influence whether they are seen as a problem or a partner.

Ultimately, this project suggests that sustainability is not about introducing entirely new ideas, but about learning how to reconnect and realign the systems that already exist. It requires working within the relationships between people, land, and water rather than attempting to control them. Sustainability depends not only on science, but on the stories, relationships, and realities of people.

In the Upper Yampa Basin, restoration presents an opportunity to heal relationships between humans and beavers by working within the realities that shape how communities interact with the landscape. This includes recognizing the stewardship already present among landowners and supporting approaches that align with their knowledge, priorities, and daily practices. Restoration requires approaching landowners with respect and offering practical support. As one rancher explained, “We don't want to change this land to make it work for us. We have to change ourselves so the land can work with us.” This perspective reflects a broader shift in how sustainability can be understood, not as control over nature, but as a relationship with it.

Beavers are one of the few species that shape their environment in ways similar to humans. This reminds us that we are not separate from the systems we manage; we are part of them. What we do shapes the landscape, and the health of that landscape directly affects us. This

creates an opportunity to move beyond short-term fixes and toward approaches grounded in patience, adaptation, and balance.

In this way, sustainability becomes less about managing resources and more about restoring relationships. When those relationships are built on trust, reciprocity, and respect, both people and ecosystems can function more effectively, creating the conditions for more resilient and connected landscapes. Within this framework, humans and beavers are not in competition. Instead, they can flourish when collaboration is chosen over control. Beavers can be dam good partners when they are treated as allies rather than something to be taken advantage of. This points toward a future where people and nature do not compete, but instead work together in a relationship grounded in reciprocity and respect.

Recommendations

Findings from this study suggest that the Upper Yampa Basin is already well-positioned to support human–beaver coexistence, with ongoing collaboration and community engagement creating momentum for future work. Building on this foundation, several key opportunities emerge.

First, increasing accessibility to educational resources is critical. Materials that blend scientific information with storytelling can make complex ecological concepts more relevant and engaging for a wider audience. *Beaver Tales* was created as a deliverable for this project, and a resource to help increase access to information about beavers and watershed health by integrating science and storytelling into an approachable format (Mefferd, 2026). The booklet was reviewed by a biologist to ensure accuracy and is available as a free resource to download and share. Printed copies are also available at the local U.S. Forest Service office to improve access for landowners and community members most likely to encounter beaver-related impacts.

Expanding the availability of similar resources can strengthen community engagement and broaden participation in coexistence efforts.

Second, targeted outreach to rural and agricultural communities is essential. Many of the individuals most directly affected by beaver activity are not consistently engaged in formal meetings or restoration discussions. Reaching these groups through place-based engagement and direct communication can help ensure that a wider range of perspectives is included in future efforts.

Third, continued investment in place-based education and experiential learning can strengthen long-term coexistence. Programs that allow participants to observe beaver systems in real-world settings, such as field tours or site visits, provide opportunities for learning through direct experience. Local organizations are already advancing this work through educational programming and community engagement, with potential to expand these efforts further.

Fourth, increasing visibility of successful coexistence in both rural and urban settings can help shift perceptions. Interpretive signage and public education efforts in areas where beavers and humans coexist can demonstrate how conflicts can be managed using available tools and strategies. For example, interpretive signage is scheduled to be installed this summer (2026) through a collaboration between Yampatika and the City of Steamboat Springs at a site where beavers are living in close proximity to human infrastructure. These visible examples can serve as accessible entry points for community learning.

Finally, this framework has potential for expansion beyond the Upper Yampa Basin. With additional time and resources, similar approaches could be applied across the broader Yampa-White-Green Basin and other watersheds. Adapting strategies to fit local ecological conditions and social contexts will be essential for success.

Figures

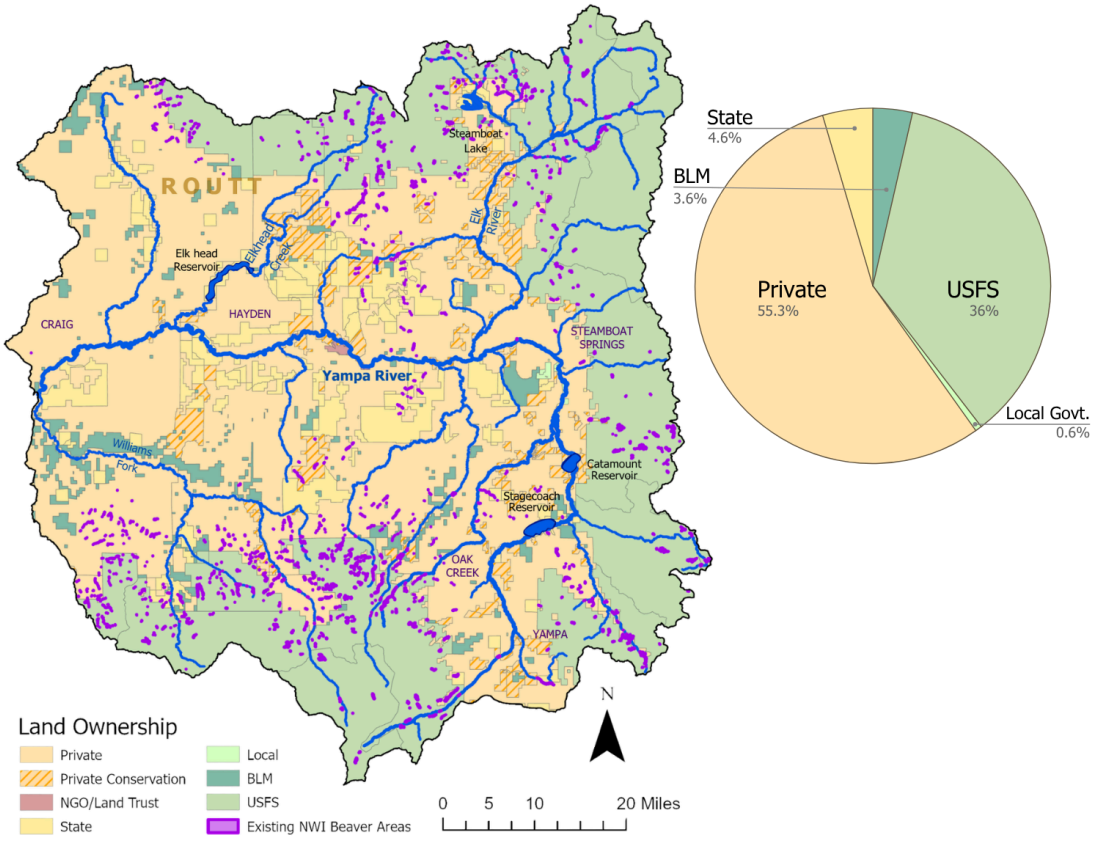


Figure 1. Upper Yampa Basin; land ownership and beaver activity

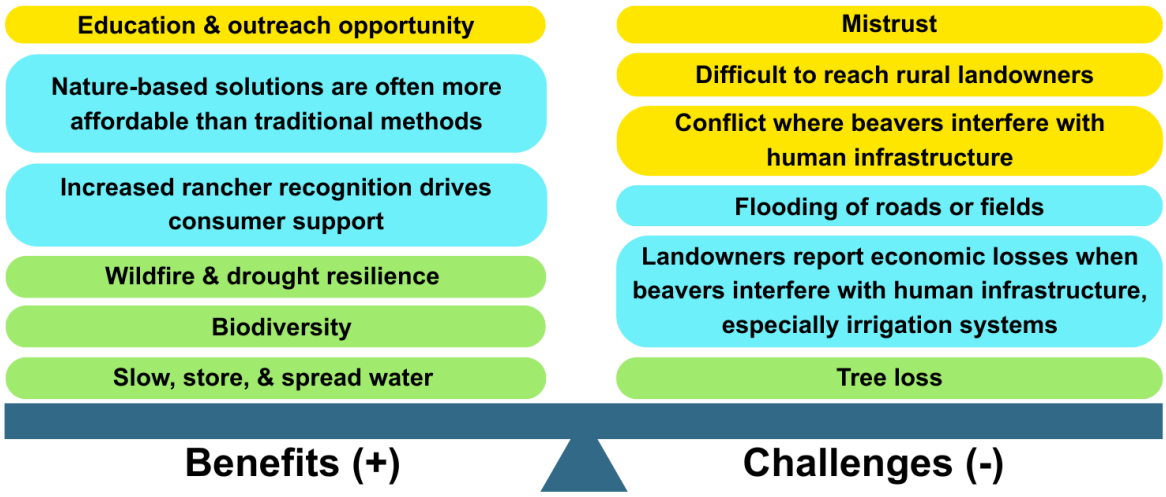


Figure 2. Weighing the Positives & Negatives; colors correspond to (figure 3) & (figure 4)



Figure 3. Word Cloud; frequently mentioned themes



Figure 4. Three Pillars of Sustainability

References

Brazier, R. E., Puttock, A., Graham, H. A., Auster, R. E., Davies, K. H., & Brown, C. M. L. (2021). *Beaver: Nature's ecosystem engineers. WIREs Water, 8*(1), e1494. https://doi.org/10.1002/wat2.1494

Breck, S. W., Wilson, K. R., & Andersen, D. C. (2001). The demographic response of bank-dwelling beavers to flow regulation: A comparison on the Green and Yampa rivers. *Canadian Journal of Zoology, 79*(11), 1957–1964. https://doi.org/10.1139/z01-162

Charnley, S. (2019). *If you build it, they will come: Ranching, riparian revegetation, and beaver colonization in Elko County, Nevada* (Research Paper PNW-RP-614). U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station. https://doi.org/10.2737/PNW-RP-614

Charnley, S., Gosnell, H., Davee, R., & Abrams, J. (2020). Ranchers and beavers: Understanding the human dimensions of beaver-related stream restoration on western rangelands. *Rangeland Ecology & Management, 73*(5), 712–723. https://doi.org/10.1016/j.rama.2020.04.008

Colorado Beaver Conservation and Management Strategy. 2026. Colorado Parks and Wildlife. Denver, CO, USA.

Erdoes, R., & Ortiz, A. (Eds.). (1984). *The gnawing* (Cheyenne legend, told by Mrs. Medicine Bull; recorded by R. Erdoes). In *American Indian myths and legends* (pp. 484–485). Pantheon Books.

Fairfax, E., & Westbrook, C. (2024). The ecology and evolution of beavers: Ecosystem engineers that ameliorate climate change. *Annual Review of Ecology, Evolution, and Systematics, 55*(1), 323–345. https://doi.org/10.1146/annurev-ecolsys-102722-122317

Fairfax, E., & Whittle, A. (2020). Smokey the beaver: Beaver-dammed riparian corridors stay green during wildfire throughout the western United States. *Ecological Applications, 30*(8), e02225. https://doi.org/10.1002/eap.2225

The Fur-Bearers. (2019). *Beavers: Coexistence strategies for municipalities and landowners.* https://thefurbearers.com/wp-content/uploads/2024-07-Beaver-Book-web.pdf

Garrity, M. (2024, February 14). *Making home: A story of beaver and babies.* Planet Forward. https://planetforward.org/story/story-beavers-babies/

Goldfarb, B. (2018). Beavers, rebooted. *Science, 360*(6393), 1058–1061. https://doi.org/10.1126/science.360.6393.1058

Goldfarb, B., & Flores, D. (2018). *Eager: The surprising, secret life of beavers and why they matter.* Chelsea Green Publishing.

Hawley-Yan, E. (2016). *Techniques for mitigating human/beaver conflicts in urban and suburban environments.* Animal Alliance of Canada.

Hodge, B. W., Lundberg, E., Jochems, L., Fields, J., Dauwalter, D. C., Neville, H., & Mayfield, M. (2025). *Process-based restoration monitoring handbook.* Trout Unlimited. https://www.researchgate.net/publication/394115645_Process-Based_Restoration_Monitoring_Handbook

Hood, G. A., Manaloor, V., & Dzioba, B. (2018). Mitigating infrastructure loss from beaver flooding: A cost–benefit analysis. *Human Dimensions of Wildlife, 23*(2), 146–159. https://doi.org/10.1080/10871209.2017.1402223

Hood, G. A., & Yarmey, N. (2015). *Mitigating human–beaver conflicts through adaptive management.* University of Alberta.

Jordan, C. E., & Fairfax, E. (2022). Beaver: The North American freshwater climate action plan. *WIREs Water, 9*(4), e1592. https://doi.org/10.1002/wat2.1592

Marvasti, A., & Treviño, A. J. (2019). *Researching social problems*. Routledge.

Meadows, D. H. (2008). *Thinking in systems: A primer.* Chelsea Green Publishing.

Meadows, D. H., Meadows, D. L., Randers, J., & Behrens, W. W. III. (1972). *The limits to growth: A report for the Club of Rome's project on the predicament of mankind.* Universe Books.

Mefferd, C. (2026). *Beaver Tales: Blending science and storytelling* [Booklet]. Colorado Mountain College. https://heyzine.com/flip-book/b34ff64d64.html

Moravek, J. A., Andruss, M., Connaughton, A., Miller, J., Garrity, M., Kirksey, K., & Fairfax, E. (2025). Using beaver capacity models: The importance of local knowledge. *Restoration Ecology, 33*(7), e70135. https://doi.org/10.1111/rec.70135

National Park Service. (2015, February 24). *Rivers and streams: Dinosaur National Monument*. https://www.nps.gov/dino/learn/nature/rivers.htm

Pollock, M. M., Lewallen, G. M., Woodruff, K., Jordan, C. E., & Castro, J. M. (Eds.). (2023). *The beaver restoration guidebook: Working with beaver to restore streams, wetlands, and floodplains* (Version 2.02). U.S. Fish and Wildlife Service. https://www.fws.gov/media/beaver-restoration-guidebook

Sherriff, L. (2021, February 23). Beaver believers: Native Americans promote resurgence of "nature's engineers." *The Guardian.* https://www.theguardian.com/environment/2021/feb/23/beavers-native-american-tribes-washington-california

Stockholm Resilience Centre [SRC]. (2025). *Planetary boundaries.* https://www.stockholmresilience.org/research/planetary-boundaries.html

Thompson, S., Vehkaoja, M., Pellikka, J., & Nummi, P. (2021). Ecosystem services provided by beavers *Castor spp. Mammal Review, 51*(1), 25–39. https://doi.org/10.1111/mam.12220

Tippie, S., & O'Brien, M. (2010). *Working with beaver for better habitat naturally.* Wildlife 2000 & Grand Canyon Trust.

Vorel, A., Dostál, T., Uhlíková, J., Korbelová, J., & Koudelka, P. (2016). *Handbook for coexisting with beavers.* Czech University of Life Sciences Prague.

Wheaton, J. M., Bennett, S. N., Bouwes, N., Maestas, J. D., & Shahverdian, S. M. (Eds.). (2019). *Low-tech process-based restoration of riverscapes: Design manual* (Version 1.0). Utah State University Restoration Consortium. http://lowtechpbr.restoration.usu.edu /manual

World Commission on Environment and Development [WCED]. (1987). *Our common future* (Report of the World Commission on Environment and Development, UN Doc. A/42/427). United Nations.